Engineered CDMA Superposition in Pretrained Transformers: Nonlinear Readout Failures, Capacity Limits, and Per-Token Surprise

Jose Joaquín Vargas Altalaguerri
Independent Researcher
jvargas8@xtec.cat

November 2026

Abstract

Sign-coded superposition can carry recoverable signals through transformer activations without providing independent computation on those signals. We examine this distinction across five application-level readouts in Qwen2.5-0.5B: normalized unembedding, attention over merged caches, multiplexed MLP probing, multi-behavior steering, and parallel output prediction. None of the tested untrained decoders gives reliable independent recovery. For example, selective CDMA cache merging increases perplexity from 8.57 to 172904, whereas matched eviction increases it by only 0.29. An exploratory width-to-symbol-information fit describes trained toy transformers, but causal-language K_{cap}^{80} ranges from 1 to 2 across evaluation samples: the largest evaluated K retaining at least 80% of the same multistream-adapted checkpoint's $K = 1$ accuracy, not intrinsic capacity. Threshold crossings are fragile; accuracy declines monotonically with stream count. Held-out target-token surprise remains descriptively associated with absolute accuracy under superposition; an additional oracle early-exit probe saves only 6.1% of layer evaluations. Modality and binding-code controls do not provide a general remedy. A paired residual-stream watermark remains recoverable at KL divergence 0.00035, providing a positive control for transport rather than parallel reasoning. These findings support a task-dependent bottleneck, not an impossibility theorem about superposition or nonlinear networks.

1 Introduction

A compact activation can retain a signal that the rest of a model cannot use as an independently addressable computation. This distinction matters whenever engineered superposition is proposed as a shortcut to batching: recovering a sign from an activation does not establish that the model processed its associated task. Our central empirical claim is that the useful superposition in these experiments lives in a linear sensing channel, or in a decoder explicitly trained for multiplexing. Each tested untrained nonlinear readout fails to turn that channel into reliable independent computation.

We develop the scientific results previously confined to the appendices of Vargas Altalaguerri [2026b]. The contribution is a controlled map of failures, their trained positive controls, and a descriptive association between target-token surprise and absolute accuracy under superposition. We do not introduce multiple-input multiple-output neural computation, already studied by Menet et al. [2023], nor claim that transformers cannot perform naturally composed tasks [Xiong et al., 2024]. "Failure at every readout" refers to the probes and operating points reported here, not every possible detector, model, or nonlinearity.

The working explanation is that task-dependent processing demand, denoted D_{task}, matters more than geometric separability alone. The causal-language experiments are consistent with this explanation, but do not identify it uniquely: optimization, adaptation budget, and the objective also differ from the toy setting. The within-sample association is heterogeneous: low-surprise targets have higher absolute accuracy under superposition, but surprise is computed retrospectively from the same predictions and is not a pre-answer routing signal.

2 The linear/nonlinear channel view

Let $h \in \mathbb{R}^d$ be a residual vector, $C \in \{-1,+1\}^{d\times B}$ a sign-code matrix, and $b \in \{-1,+1\}^B$ a payload. A normalized additive signal is

$$h' = h + \delta, \qquad \delta = \alpha\|h\|_2 \frac{Cb}{\|Cb\|_2}, \qquad \widehat{b} = \operatorname{sign}\Big(C^\top(h' - h)\Big). \tag{1}$$

The implementation scales the columns of C by $d^{-1/2}$, which cancels in the normalized injection. For task multiplexing, the analogous construction binds each stream x_k coordinatewise:

$$z = \sum_{k=1}^{K} c_k \odot E(x_k), \qquad \widehat{x}_k = G(c_k \odot F(z)). \tag{2}$$

Here E is an embedding, F the shared computation, and G the readout. A linear map transports additive components exactly, but need not preserve code orthogonality or signal-to-noise ratio. Training can adapt E, F, G to their binding scheme.

For a differentiable nonlinear map f, local transport is instead

$$f(h+\delta) - f(h) = J_f(h)\delta + \mathcal{O}(\|\delta\|_2^2). \tag{3}$$

The Jacobian depends on the operating point; a fixed correlator designed for C need not match $J_f(h)C$. Attention additionally changes the routing weights of the mixture. Neither observation proves information destruction. In particular, the unembedding matrix itself is linear: "logit barrier" below includes the intervening stack, final RMS normalization, and a restricted detector, not a nonlinear matrix multiplication.

We use D_{task} as a conceptual demand on nonlinear processing. The heuristic $KD_{\text{task}} \lesssim C_{\text{nl}}(d)$ motivates the experiments; neither quantity is independently measured. The toy relation to $\log_2 V$ is an empirical fit, not a general channel-capacity theorem or a bound on inference speed.

3 Experimental setup

The pretrained-model experiments use Qwen2.5-0.5B. Its recorded width is $d = 896$ and depth is 24 layers. The real-language and selective-multiplexing probes adapt this checkpoint on WikiText; other probes test fixed pretrained weights. The toy models use synthetic sequence reversal or masked token reconstruction, with geometry and training settings stored alongside their results. These objectives must not be conflated with autoregressive language modeling.

Accuracy is exact target-token agreement, perplexity is exponentiated mean negative log likelihood, and watermark quality is bit agreement. Output fidelity is $D_{\text{KL}}(p_{\text{clean}}\|p_{\text{marked}})$ and greedy-token agreement. Raw, relative, and chance-subtracted capacity criteria differ:

$$K_{\text{abs}} = \max\{K : A(K) \geq \tau_{\text{abs}}\}, \tag{4}$$

$$K_{\text{rel}} = \max\{K : A(K)/A(1) \geq \tau_{\text{rel}}\}, \tag{5}$$

$$K_{\text{signal}} = \max\{K : [A(K) - q]/[A(1) - q] \geq \tau_{\text{rel}}\}, \tag{6}$$

where q is the chance baseline. The toy absolute threshold is 0.90; the relative-threshold definition is encoded in the producer's `rel80`/`signal80` fields. Capacity is the largest qualifying *tested* point. At the sweep endpoint it is a lower bound, and it is not meaningful if the single-stream signal is negligible.

All reported measurements are regenerated from frozen JSON. Watermark sample counts are not stored in that JSON; its producer specifies the prompt/position protocol rather than the smaller prompt count in the earlier manuscript. Positions from the same prompt are correlated; aggregate results do not support independent-sample confidence intervals. Clean and marked activations/logits are subtracted exactly, position by position. Divergent continuations also change context, so later differences are not pure local perturbations.

4 Five nonlinear readout failures

4.1 Normalized unembedding and logits

Table 1 contains every recorded amplitude and payload size, including the random-bit control. For the smallest payload and amplitude, residual recovery is 0.9194 while the transported-code logit detector reaches 0.5684. The latter is weakly above nominal chance, so "at chance" would overstate the result without a valid uncertainty estimate. Larger amplitudes strongly perturb output without producing high-fidelity logit recovery.

The detector correlates paired logit differences with WC, where W is the unembedding matrix. It omits the exact final-normalization Jacobian and does not fit an inverse or nonlinear decoder. Thus the result rejects this simple detector as a useful readout; it does not show that logits contain no recoverable information. Access to paired logits also exceeds ordinary text-only access.

Table 1: Paired watermark readouts: every recorded payload size and amplitude.

B	α	Residual	Logits	Control	KL	Agreement
16	0.01	0.9194	0.5684	0.5020	0.00035	1.0000
16	0.02	0.9170	0.5581	0.5034	0.06077	0.9766
16	0.05	0.8774	0.5488	0.4980	1.18703	0.8281
16	0.1	0.8262	0.5620	0.4966	2.45799	0.6484
16	0.2	0.7944	0.5386	0.5039	3.95222	0.4141
16	0.4	0.8569	0.5322	0.5020	5.92412	0.1484
64	0.01	0.7717	0.5480	0.4966	0.00029	0.9922
64	0.02	0.7704	0.5459	0.4987	0.00117	0.9922
64	0.05	0.7308	0.5331	0.4957	1.64037	0.8203
64	0.1	0.7002	0.5289	0.4958	3.57792	0.6094
64	0.2	0.6914	0.5305	0.4939	3.90638	0.4375
64	0.4	0.7450	0.5432	0.5028	5.10210	0.2734
256	0.01	0.6435	0.5220	0.4977	0.00032	0.9922
256	0.02	0.6385	0.5221	0.4981	0.43778	0.9609
256	0.05	0.6361	0.5218	0.4973	0.73892	0.8984
256	0.1	0.6268	0.5249	0.4977	1.48889	0.7969
256	0.2	0.5987	0.5185	0.5001	4.44739	0.3906
256	0.4	0.6246	0.5251	0.5013	5.47730	0.2188

Exact clean–marked subtraction; the logit detector projects onto transported codes, not an optimized decoder.

4.2 Attention softmax and selective KV merging

The cache experiment selects low-surprise positions and compares CDMA merging, arithmetic-mean merging, eviction, and a random-position CDMA control at matched storage reduction. It preserves absolute positional indexing. Table 2 shows catastrophic CDMA merging, while eviction yields an effective-context multiplier of 1.50 at a perplexity increment of 0.29. Arithmetic averaging also substantially outperforms sign-coded merging. This isolates failure of the proposed cache representation rather than failure of all cache compression.

Table 2: Complete KV-cache comparison at the recorded compression setting.

Method	NLL	PPL	Removed fraction	K_{eff}
FULL	2.149	8.572	0.00000	1.000
CDMA-MERGE-EASY	12.060	172903.667	0.33301	1.499
MEAN-MERGE-EASY	2.271	9.692	0.33301	1.499
EVICT-EASY	2.181	8.858	0.33301	1.499
CDMA-MERGE-RANDOM	12.320	224130.095	0.33301	1.499

K_{eff} denotes effective context per retained slot, not the number of CDMA streams. No sweep was recorded.

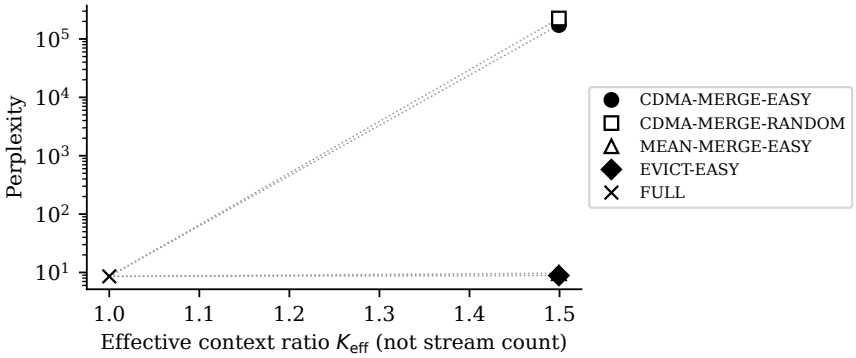


Figure 1: KV perplexity versus effective context ratio. Only the full-cache baseline and one compressed operating point exist; dotted connectors are not measured curves. The horizontal variable is not CDMA stream count.

4.3 MLP nonlinearity and "neurosurgery"

A targeted set of 116 MLP neurons removes the measured concept response, but also damages general perplexity (Table 3). Unrelated accuracy is insufficient to certify capability preservation: the narrow control remains intact while fluency deteriorates. Random ablations of the same size are markedly less destructive to perplexity. The multiplexed concept scan also fails to reproduce its serial maps (Table 4). These are application-level failures, not isolated causal measurements of the activation function.

Table 3: MLP ablation: targeted versus matched-size random control.

Fraction (%)	Neurons	Target acc.	Other acc.	Target PPL	Random acc.	Other acc.	Random PPL
Baseline	–	0.400	1.000	12.180	–	–	–
0.1	116	0.000	1.000	18.290	0.362	1.000	12.240
0.5	583	0.000	1.000	24.230	0.476	1.000	12.370
1.0	1167	0.000	1.000	28.770	0.505	1.000	12.770
2.0	2334	0.000	1.000	34.700	0.514	1.000	13.170
5.0	5836	0.000	1.000	44.560	0.448	1.000	15.730

Table 4: Multiplexed versus serial neuron maps, by concept.

Concept	Top-list overlap	Cosine
capital	0.260	-0.196
chemistry	0.340	-0.127
continent	0.240	-0.162
antonym	0.360	-0.155
Mean	0.300	-0.160

4.4 Downstream stack and eigen-steering

Orthogonalizing behavior vectors reduces mean off-target coupling by only 2.15%, while reducing mean on-target retention by 37.9% relative to naive addition. The previous shorthand "halves retention" is not numerically exact. Table 5 reports every behavior, single-vector baseline, simultaneous condition, and random control. Ratios may be negative or exceed unity, so their mean must not be interpreted as a bounded success probability. Better input geometry does not ensure independent downstream behavioral control.

4.5 Output and parallel multiplexing

Table 6 compares extracted and random directions at every recorded stream count and amplitude. Hidden-state correlation can remain high while the original per-stream output targets do not survive.

Table 5: All steering behaviors and controls; effects are changes from baseline.

Behavior	Single	Naive	Ortho.	Random	R_{naive}	R_{ortho}
sentiment	0.067	-0.067	-0.067	-0.067	-1.000	-1.000
language	1.127	1.013	1.073	0.000	0.899	0.953
verbosity	0.130	0.230	0.137	-0.970	1.769	1.051
topic_food	0.400	0.033	0.033	0.000	0.083	0.083
Mean retention	–	–	–	–	0.438	0.272
Mean cross-coupling	–	0.387	0.379	–	–	–

Retention is a ratio to the single-behavior effect: negative values and values above unity are possible.

This is a transport–computation dissociation, not evidence of practical parallel generation. The additional sensing and depth controls are retained in Tables 21–22; the stored depth-break statistic has a pre-injection numerical artifact and is not used as a quantitative depth limit.

Table 6: Output multiplexing: all amplitudes, stream counts, and both code controls.

α	K	h_E	n_E	s_E	t_E	h_R	n_R	s_R	t_R
0.050	2	1.000	1.000	0.000	1.000	1.000	0.052	0.000	1.000
0.050	4	1.000	1.000	0.000	0.750	1.000	0.103	0.000	1.000
0.050	8	1.000	1.000	0.000	0.880	1.000	0.202	0.000	1.000
0.050	16	1.000	1.000	0.000	0.940	1.000	0.558	0.000	0.940
0.050	32	0.938	1.000	0.000	0.940	1.000	0.547	0.000	1.000
0.150	2	1.000	1.000	0.000	1.000	1.000	0.048	0.000	1.000
0.150	4	1.000	1.000	0.000	1.000	1.000	0.230	0.000	1.000
0.150	8	1.000	1.000	0.000	1.000	1.000	0.491	0.000	1.000
0.150	16	1.000	1.000	0.000	0.940	0.875	0.651	0.000	1.000
0.150	32	0.750	1.000	0.000	0.910	0.625	0.834	0.000	0.970
0.300	2	1.000	1.000	0.000	1.000	1.000	0.312	0.000	1.000
0.300	4	1.000	1.000	0.000	1.000	1.000	0.368	0.000	1.000
0.300	8	1.000	1.000	0.000	1.000	1.000	0.456	0.000	1.000
0.300	16	0.750	1.000	0.000	1.000	1.000	0.563	0.000	1.000
0.300	32	0.813	1.000	0.031	0.940	0.937	0.669	0.000	0.940

E/R: extracted/random; h: hidden correlation; n: null ratio; s: logit survival; t: distinct-target fraction.

5 Exploratory empirical capacity fit

5.1 Repaired toy configurations

The repaired synthetic reversal cells yield an exploratory empirical capacity fit

$$K_{\text{cap}} \approx a \, \frac{d}{\log_2 V}, \qquad a \simeq 0.37. \tag{7}$$

The through-origin least-squares coefficient is 0.377; 0.37 is its truncated reference. Repaired cells replace matching original configurations, and the fit uses that substituted set, not an independent validation set (Table 7). This is not a capacity law: validation on configurations and seeds excluded from fitting has not been performed. Complete accuracy grids appear in Table 18.

5.2 LoRA, full fine-tuning, and rank

The toy rank sweep plateaus at 6 streams against a full-control capacity of 14 (Table 8). The adaptation-location ablation (Table 9) shows that parameterization matters in the toy regime. We do not fit a logarithmic rank relation from this sparse grid.

The original LoRA and full-fine-tuning curves are *in-sample*: evaluation resamples training tokens. Their threshold crossing is not intrinsic capacity: K_{cap}^{80} is the largest evaluated K retaining at least 80% of the $K = 1$ accuracy of the same checkpoint trained with a multistream curriculum, not a separately adapted single-stream model. The full/initial LoRA grid is $K \in \{1, 2, 4, 6, 8\}$; the

Table 7: Every toy capacity cell, before and after targeted curriculum repairs.

Run	d	V	K_{cap}	$K_{\text{cap}} \log_2(V)/d$
Original	64	16	0	0.000
Original	128	16	12	0.375
Original	256	16	24	0.375
Original	64	64	0	0.000
Original	128	64	4	0.188
Original	256	64	16	0.375
Repair	64	16	5	0.312
Repair	64	64	3	0.281
Repair	128	64	10	0.469

The fit uses repaired cells in place of their original counterparts. Original failures remain visible; repairs change the training curriculum/budget.

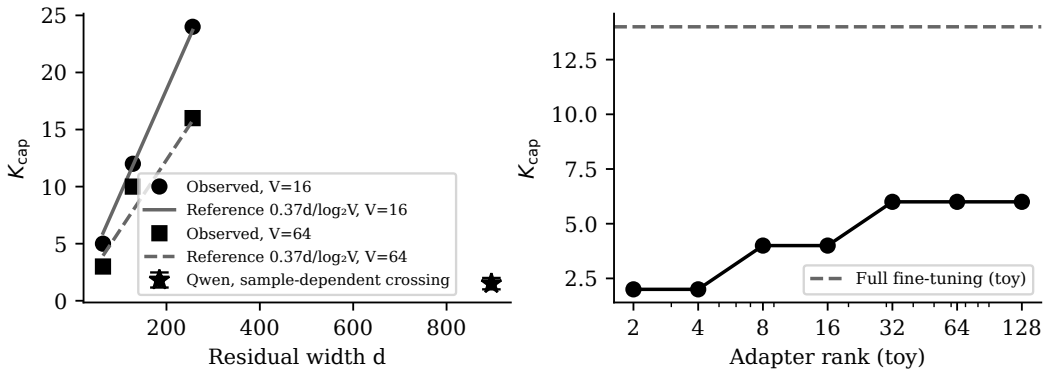


Figure 2: Left: repaired toy capacity and exploratory fit reference; the Qwen marker spans observed train/held-out threshold crossings on $K \in \{1, 2, 4, 6, 8\}$, one checkpoint and one budget, retaining at least 80% of its $K = 1$ accuracy, not an intrinsic capacity. Right: toy adapter-rank sweep and full-training control. No real-language vocabulary size is recorded in these JSON, so a numerical extrapolation to Qwen is not fabricated.

Table 8: Complete adapter-rank sweep on the toy task.

Rank	Trainable parameters	K_{cap}	1	2	4	6	8	10	12	14	16
2	15888	2	0.997	0.949	0.472	0.196	0.089	0.074	0.072	0.069	0.068
4	29712	2	1.000	0.997	0.735	0.415	0.106	0.083	0.078	0.073	0.070
8	57360	4	1.000	1.000	0.937	0.629	0.129	0.098	0.087	0.082	0.077
16	112656	4	0.992	1.000	0.994	0.837	0.246	0.123	0.107	0.096	0.091
32	223248	6	1.000	1.000	0.999	0.950	0.472	0.163	0.144	0.129	0.118
64	444432	6	1.000	1.000	0.999	0.978	0.762	0.177	0.157	0.144	0.128
128	886800	6	0.993	0.998	0.998	0.977	0.816	0.192	0.170	0.154	0.139
Full	613776	14	–	–	–	–	–	–	–	–	–

Accuracy columns are indexed by stream count. The full-control JSON records capacity but not its curve.

Table 9: Toy adaptation ablation, with complete accuracy curves.

Condition	1	2	4	6	8	10	12	14	16	K_{cap}
full	1.000	1.000	1.000	1.000	1.000	0.995	0.978	0.935	0.844	14
lora_both	1.000	1.000	0.987	0.777	0.192	0.123	0.108	0.099	0.093	4
lora_mlp	0.999	0.998	0.704	0.254	0.080	0.070	0.066	0.066	0.065	2
lora_attn	1.000	0.998	0.852	0.489	0.162	0.121	0.109	0.102	0.095	2
frozen	1.000	0.193	0.104	0.081	0.072	0.069	0.066	0.066	0.064	1

selective LoRA evaluation omits the intermediate full-FT grid point. Each adaptation uses one checkpoint and one training budget. Table 10 now includes independently evaluated WikiText validation tokens. Full fine-tuning gives new-train $K = 1$ accuracy 0.246 and held-out accuracy 0.207, with threshold crossings 1 and 2, respectively. The selective LoRA run similarly crosses at 1 on new train windows and 2 held-out. Thus K_{cap}^{80} lies between 1 and 2 depending on evaluation sample, rather than establishing a fixed language limit. For example, full-FT train retention at $K = 2$ is 76.8%, below the 80% cutoff; held-out retention is 87.0%. The threshold statistic is fragile near that cutoff. The robust pattern is monotonically declining absolute accuracy: $K = 2$ retains 76.8–87.0% of $K = 1$ accuracy, while $K = 8$ retains 39.6–51.2% across the new train/held-out curves (mean 45.0%, approximately half). These aggregate results do not supply uncertainty estimates or exclude optimization/budget effects.

Table 10: Causal language modeling: original in-sample curves and train/held-out evaluation.

Condition	1	2	4	6	8	K_{cap}
No adaptation / original train	0.007	0.008	0.013	0.008	0.005	–
LoRA / original train	0.219	0.198	0.144	0.115	0.088	2
Full FT / original train	0.244	0.215	0.159	0.135	0.113	2
Full FT / new train	0.246	0.189	0.157	0.131	0.109	1
Full FT / held-out	0.207	0.180	0.154	0.137	0.106	2
LoRA selective / new train	0.235	0.183	0.147	–	0.093	1
LoRA selective / held-out	0.210	0.175	0.146	–	0.094	2

Train rows are in-sample; held-out rows are out-of-sample. The denominator is K=1 accuracy of the same multistream-adapted checkpoint, not a separately adapted model. Capacity is a grid threshold crossing, not an intrinsic limit.

6 Descriptive association: target-token surprise and absolute accuracy

For a target token y_t, define surprise under the adapted single-stream model as

$$S_t = -\log p_1(y_t \mid x_{<t}), \qquad R_g(K) = \frac{A_g(K)}{A_g(1)}, \tag{8}$$

where g indexes a fixed surprise tercile. The original numbers are *in-sample*, computed on the training-token pool. Cut points are 3.13 and 7.57 nats; easy-token accuracy at $K = 1$ is 0.597, versus constant-token-prior accuracy 0.124. The hard tercile has zero correct targets at $K = 1$, so its relative retention is undefined. Table 11 distinguishes these train results from any new split evaluation. The held-out association is now supported out of sample: easy-token accuracy at $K = 1, 2, 4, 8$ is 61.8, 50.0, 40.6, and 25.1%; easy-token retention is 100.0, 80.9, 65.7, and 40.6%. The middle group has low absolute accuracy and the hard group remains at zero even at $K = 1$ (Table 11). Compared with new-train easy-token $K = 1$ accuracy 0.686, held-out accuracy is 0.618; the qualitative association persists, while the overall threshold crossing changes as discussed above. Both splits use 32 evaluation batches per stream count. Terciles are fitted independently within each evaluated split, not trained on one and transferred to the other; held-out cut points are 3.10 and 7.79 nats. The shared-prediction coupling still applies. Surprise and correctness are computed from the same single-stream predictions, so the association is descriptive and baseline enrichment is expected. The result concerns *absolute* target-token accuracy under superposition; relative retention does not order the groups: at $K = 4$, $R_E = 0.698$ whereas $R_M = 2.154$, with an unstable small middle-tercile denominator. Target-token surprise requires the answer and is not a predictor available before the response. No selective scheduler or throughput saving is established.

A negative early-exit control. The additional probe applies the final normalization and output head at each layer and finds the earliest layer after which all argmax predictions agree with the final one. Across 1024 positions, mean exits range from 21.08 to 23.68 of 24 layers, and the pooled

Table 11: Absolute target-token accuracy and relative retention by surprise tercile, with available split evaluation.

Split	K	Easy	Middle	Hard	All	R_E	R_M	R_H
Original train	1	0.597	0.013	0.000	0.203	1.000	1.000	–
Original train	2	0.506	0.018	0.000	0.175	0.848	1.385	–
Original train	4	0.417	0.028	0.000	0.148	0.698	2.154	–
Original train	8	0.265	0.024	0.000	0.096	0.444	1.846	–
Original train	Prior	0.124	0.008	0.000	0.044	–	–	–
New train	1	0.686	0.019	0.000	0.235	1.000	1.000	–
New train	2	0.511	0.037	0.000	0.183	0.745	1.947	–
New train	4	0.402	0.039	0.000	0.147	0.586	2.053	–
New train	8	0.246	0.031	0.000	0.093	0.359	1.632	–
New train	Prior	0.117	0.013	0.000	0.043	–	–	–
Held-out	1	0.618	0.010	0.000	0.210	1.000	1.000	–
Held-out	2	0.500	0.026	0.000	0.175	0.809	2.600	–
Held-out	4	0.406	0.033	0.000	0.146	0.657	3.300	–
Held-out	8	0.251	0.030	0.000	0.094	0.406	3.000	–
Held-out	Prior	0.131	0.011	0.000	0.047	–	–	–

Train rows are in-sample; held-out rows use validation tokens. Terciles are fitted independently within each split, not transferred from train. Zero-denominator retention is undefined (–); relative retention need not order the terciles.

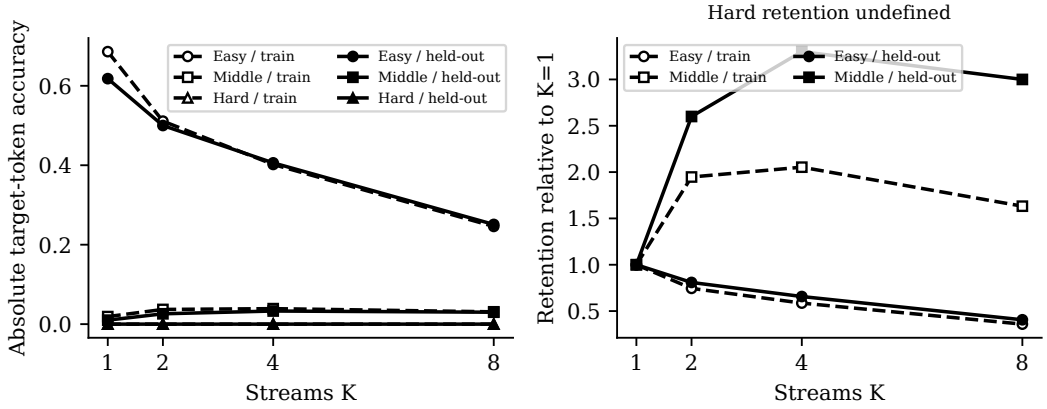


Figure 3: New train and held-out absolute accuracy and relative retention by target-token surprise tercile, with split-specific terciles. Relative retention does not order the groups; the middle-tercile ratio has a small denominator and hard-tercile retention is undefined.

layer-count saving is only 6.1% (Table 12). This is an oracle stable-exit statistic, not an executable stopping policy. It excludes head-evaluation overhead and does not measure latency. Its surprise is final-distribution top-prediction surprise, unlike the target-token surprise above. Therefore the result is consistent with limited depth redundancy, but not a direct test of the same surprise partition. Heterogeneity in recoverability does not automatically imply early computation can be skipped.

Table 12: Oracle stable-argmax early-exit probe (no trained exit heads).

Group	Mean exit layer	Exit by half depth (%)	Layer-count saving (%)
Easy	21.08	0.00	12.20
Middle	22.85	0.00	4.80
Hard	23.68	0.00	1.30
All	22.54	0.00	6.10

Savings exclude exit-head cost and are not measured wall-clock or FLOP reductions; surprise differs from the target-token accuracy probe.

7 Modality and code design are not general levers

The masked-reconstruction controls separate task and tokenizer effects from causal-language adaptation. Image capacity depends on vocabulary and setup; it is not a universal modality constant. With the larger recorded vocabulary, k-means and VQ-VAE image tokens have capacity 4, whereas text and code remain signal-preserving to the extended sweep endpoint of 32 streams. The latter is a lower bound, not an observed failure threshold. Figure 4 distinguishes censored endpoints; Tables 13–16 retain every measured accuracy. Hadamard, random-sign, and highly correlated bind-

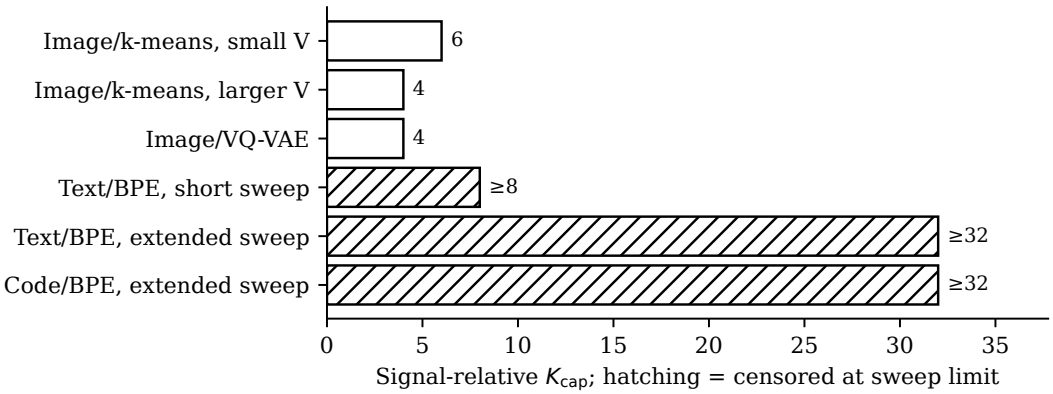


Figure 4: Signal-relative capacity across modality/tokenizer controls. Hatching denotes a sweep-limited lower bound. These masked toy objectives are not matched to causal language modeling.

ing codes all preserve reconstruction through the tested endpoint under training (Table 17). This refutes code orthogonality as a sufficient practical lever in this trained setting, not its relevance in every model. It is consistent with learned embeddings compensating for the code, but does not prove that mechanism. The results alone also do not identify image-token rank, correlation, or redundancy as the cause of its lower capacity.

Table 13: Modality / tokenizer comparison: vision_kcap.

Condition / state	1	2	3	4	6	8	K_{cap}	Signal at single stream
random / untrained	0.0150	0.0150	0.0160	0.0160	0.0160	0.0160	–	–
random / trained	0.0150	0.0160	0.0150	0.0160	0.0160	0.0150	1	-0.0006
text / untrained	0.0010	0.0350	0.0300	0.0310	0.0210	0.0190	–	–
text / trained	0.1950	0.1910	0.1920	0.1930	0.1930	0.1920	8	0.1794
image / untrained	0.0110	0.0130	0.0190	0.0160	0.0170	0.0150	–	–
image / trained	0.3450	0.3310	0.3260	0.3240	0.3000	0.2640	6	0.3294

Capacity is signal-relative. A capacity at the largest tested stream count is a lower bound; random controls have negligible signal and no meaningful capacity.

Table 14: Modality / tokenizer comparison: vision_kcap_hardvocab.

Condition / state	1	2	3	4	6	8	K_{cap}	Signal at single stream
random / trained	0.0019	0.0019	0.0022	0.0020	0.0019	0.0022	1	-0.0001
text_bpe / trained	0.3723	0.3669	0.3650	0.3620	0.3600	0.3582	8	0.3703
image / trained	0.1996	0.1916	0.1732	0.1687	0.1465	0.1145	4	0.1976

Capacity is signal-relative. A capacity at the largest tested stream count is a lower bound; random controls have negligible signal and no meaningful capacity.

8 Positive control: the residual-stream watermark survives

At $B = 16$ and $\alpha = 0.01$, paired residual recovery is 0.9194, with KL 0.00035 and greedy-token agreement 1.0000 (Table 1). Figure 5 contrasts this with the weak logit detector. The small but

Table 15: Modality / tokenizer comparison: vqvae_kcap.

Condition / state	1	2	3	4	6	8	K_{cap}	Signal at single stream
image_vqvae / trained	0.1011	0.1008	0.0928	0.0932	0.0787	0.0631	4	0.0991
image_kmeans / trained	0.1996	0.1916	0.1732	0.1687	0.1465	0.1145	4	0.1976
text_bpe / trained	0.3723	0.3669	0.3650	0.3620	0.3600	0.3582	8	0.3703
random / trained	0.0019	0.0019	0.0022	0.0020	0.0019	0.0022	1	-0.0001

Capacity is signal-relative. A capacity at the largest tested stream count is a lower bound; random controls have negligible signal and no meaningful capacity.

Table 16: Modality / tokenizer comparison: code_kcap.

Condition / state	1	2	4	8	12	16	24	32	K_{cap}	Signal at single stream
random / trained	0.0022	0.0023	0.0016	0.0020	0.0019	0.0020	0.0020	0.0020	1	0.0002
text_bpe / trained	0.3571	0.3540	0.3637	0.3613	0.3556	0.3609	0.3567	0.3577	32	0.3551
code / trained	0.3185	0.3244	0.3244	0.3229	0.3236	0.3216	0.3249	0.3221	32	0.3165

Capacity is signal-relative. A capacity at the largest tested stream count is a lower bound; random controls have negligible signal and no meaningful capacity.

Table 17: Modality / tokenizer comparison: coded_vocab_kcap.

Condition / state	1	2	4	8	16	32	K_{cap}	Coherence
random_pm1 / trained	0.3562	0.3633	0.3534	0.3607	0.3597	0.3588	32	0.2656
hadamard / trained	0.3715	0.3623	0.3589	0.3529	0.3586	0.3584	32	0.0000
correlated / trained	0.3479	0.3636	0.3567	0.3573	0.3582	0.3586	32	0.9687

Capacity is signal-relative. A capacity at the largest tested stream count is a lower bound; random controls have negligible signal and no meaningful capacity.

nonzero divergence should not be called exact invariance. This positive control supports recoverable low-amplitude activation transport, not a robust text watermark or independent task computation. The companion *Paper A: Activation-side session tagging for LLM audit logs* [Vargas Altalaguerri, 2026a], by the same author in the same repository, treats the audit-log system; it has no assigned DOI. Its scale study (Qwen2.5-0.5B/1.5B/7B, 128 sessions of 128 greedy tokens) confirms the linear-channel picture in both directions: the tag is recovered only by paired replay against the logged residual, blind decoding with a static calibration is at chance, a tag added directly at the read layer decodes perfectly (the residual response to a small perturbation is close to identity, so the mark proves transport, not computation), and the same α that leaves next-token agreement at 1.0000 here changes most 128-token continuations on the larger models. The present exact paired-subtraction experiment must not be equated with a deployable single-log verifier. Trained MLP and

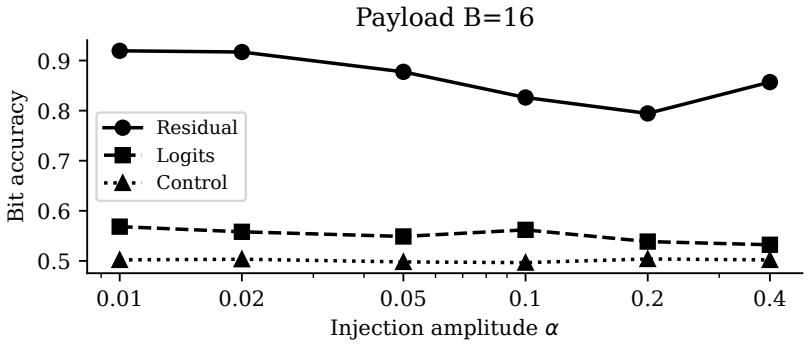


Figure 5: Residual, logit, and random-control bit accuracy for the smallest recorded payload, over all injection amplitudes.

transformer controls (Tables 20 and 19) demonstrate that nonlinear networks can learn multiplexing. They constrain the failure claim to untrained engineered readouts and their objectives.

9 Related work and honest novelty

Elhage et al. [2022] study learned feature superposition, not imposed codes. Menet et al. [2023] train multiple-input multiple-output networks for superposed computation. Natural in-context task composition [Xiong et al., 2024] differs from sign-bound activation addition. ParScale [Chen et al., 2025] aggregates separate learned parallel passes, not one unmodified residual mixture.

Borobia et al. [2026] study leverage dispersion and near-floor geometry in trained overcomplete codes, not a theorem about the nonlinear application failures tested here.

EntropyCache [Cheong et al., 2026] uses decoded-token entropy for diffusion-model cache refresh, not causal-model multiplexing. Text watermarking [Kirchenbauer et al., 2023] changes token selection, unlike paired activation sensing. BitNet [Wang et al., 2023] and Mamba [Gu and Dao, 2023] change weight computation or sequence architecture, without establishing free capacity for CDMA streams. Our contribution is the combined negative evidence and its surprise-stratified qualification, not a new multiplexing mechanism or universal information-theoretic law.

10 Limitations

All real-LLM results concern one small checkpoint, without cross-family or larger-scale replication. The original adaptation and surprise accuracies measure within-corpus recovery; the new validation-token evaluations provide out-of-sample evidence for these trained runs, not independent-seed or cross-model replication. Sample sizes are small and correlated, seeds are not comprehensively replicated, and rounded aggregate JSON does not support reliable uncertainty estimates. Behavioral scores are narrow. The logit probe tests a restricted correlator, not optimal or text-only decoding, and the application probes do not causally isolate individual nonlinear operators.

The exploratory capacity fit comes from toy reversal with replacement training configurations, without validation on unseen configurations or seeds. Masked modality tasks, causal language modeling, and MLP sign recovery use different objectives and thresholds. Endpoint capacities are censored. Matching in-sample language threshold crossings under the tested budgets do not establish an intrinsic D_{task} limit. Surprise is partly tied to baseline correctness, and the early-exit statistic is retrospective. No scheduler, wall-clock acceleration, or general impossibility result follows from these experiments.

Changes from version 1

We correct arXiv:2605.01192 and its earlier floor-theorem attribution, omit the unsupported real-language overprediction, call the toy relation exploratory, and report original in-sample and new held-out evidence separately, including the evaluation-dependent threshold crossing. The readout failures are not impossibility results.

Reproducibility

Code and frozen results: `https://github.com/ttzrs/neural-cdma`. The previous manuscript is DOI 10.5281/zenodo.20570543, not a DOI for this paper. From `papers/channel/`, run `bash build.sh`; no GPU or weights are required. `tables/provenance.json` records source hashes and numeric-macro derivations.

Readouts: `watermark_provenance.py`, `kv_selective.py`, `cdma_neurosurgery.py`, `eigen_steering.py`, `multithread.py`.

Capacity: `capacity_law.py`, `capacity_fix.py`, `rank_sweep.py`, `hybrid_multiplex.py`, `validate_llm*.py`, `validate_llm_full_heldout.py`.

Surprise: `prototype_selective.py`, `prototype_selective_heldout.py`, `early_exit_surprise.py`.

Modality: `vision_kcap*.py`, `vqvae_kcap.py`, `code_kcap.py`, `coded_vocab_kcap.py`.

Controls: `train_superposition.py`, `train_transformer_superposition.py`, `viability.py`, `profile_depth.py`.

Rebuilding regenerates presentation, not experiments.

A Complete supporting controls

These tables retain toy accuracy grids, trained sign-accuracy controls, sensing results, and depth summaries. Raw regression errors and depth arrays remain in the hashed JSON; the break statistic is not used as a usable-depth claim.

Table 18: Complete toy accuracy curves (missing grid points are not interpolated).

Condition	1	2	3	4	5	6	8	10	12	16	20	24	28	32	K_{cap}
Original d=64,V=16	0.271	0.278	–	0.249	–	0.218	0.198	–	0.162	0.137	–	–	–	–	0
Original d=128,V=16	1.000	1.000	–	1.000	–	1.000	0.996	–	0.935	0.804	0.655	0.474	0.299	0.219	12
Original d=256,V=16	1.000	1.000	–	1.000	–	1.000	1.000	–	1.000	0.999	0.989	0.958	0.897	0.793	24
Original d=64,V=64	0.187	0.176	–	0.129	–	0.101	0.084	–	0.063	0.052	–	–	–	–	0
Original d=128,V=64	0.974	0.974	–	0.941	–	0.806	0.626	–	0.274	0.154	0.107	0.086	0.070	0.063	4
Original d=256,V=64	1.000	1.000	–	1.000	–	1.000	0.999	–	0.983	0.923	0.830	0.729	0.633	0.544	16
Repair d=64,V=16	1.000	1.000	0.997	0.990	0.954	0.873	0.614	0.345	–	–	–	–	–	–	5
Repair d=64,V=64	0.994	0.997	0.974	0.894	0.757	0.548	–	–	–	–	–	–	–	–	3
Repair d=128,V=64	1.000	1.000	–	1.000	–	0.997	0.972	0.900	0.773	–	–	–	–	–	10

Table 19: Trained-transformer positive control: token-reversal accuracy.

Condition	1	2	3	4	6	8	K_{cap}
Untrained	0.057	0.061	0.064	0.062	0.062	0.065	–
Trained	1.000	0.999	0.998	0.988	0.967	0.860	–

Table 20: Trained-MLP controls: sign accuracy for every recorded stream count.

Task / codes	State	1	2	4	8	16	24	32	48
identity_fixedC	untrained	0.494	0.514	0.505	0.503	0.506	0.498	0.498	0.505
identity_fixedC	trained	0.997	0.984	0.967	0.946	0.920	0.898	0.887	0.857
identity_learnedC	untrained	0.501	0.489	0.495	0.502	0.498	0.500	0.499	0.495
identity_learnedC	trained	0.999	0.979	0.960	0.945	0.917	0.895	0.882	0.854
square_fixedC	untrained	0.517	0.514	0.501	0.512	0.509	0.508	0.508	0.508
square_fixedC	trained	0.958	0.951	0.934	0.902	0.859	0.821	0.800	0.766
square_learnedC	untrained	0.538	0.524	0.536	0.514	0.519	0.510	0.513	0.510
square_learnedC	trained	0.970	0.959	0.940	0.917	0.893	0.865	0.847	0.818

These sign-accuracy probes are not token-accuracy capacity measurements; regression errors remain available in the source JSON.

Table 21: Linear-channel sensing sweep; extracted (E), Gold-like (G), and random (R) codes.

K	A_E	r_E	g_E	D_E	A_G	r_G	g_G	D_G	A_R	r_R	g_R	D_R
1	1.000	0.000	3.466	0.377	1.000	0.000	0.745	0.027	1.000	0.000	0.438	0.016
2	1.000	1.000	2.160	0.194	1.000	0.000	0.615	0.026	1.000	0.000	0.543	0.050
4	1.000	0.944	1.472	0.510	1.000	0.993	0.524	0.032	1.000	0.993	0.475	0.028
8	1.000	0.873	1.233	1.514	1.000	0.989	0.421	0.086	1.000	0.956	0.603	0.076
16	1.000	0.920	0.770	0.382	1.000	0.985	0.471	0.026	1.000	0.954	0.465	0.063
32	1.000	0.918	0.997	0.318	1.000	0.957	0.510	0.034	1.000	0.936	0.525	0.020
64	1.000	0.957	0.621	0.199	1.000	0.940	0.443	0.056	1.000	0.928	0.625	0.031
128	0.992	0.945	0.536	0.175	0.891	0.779	0.734	0.015	0.984	0.899	0.488	0.033
256	0.965	0.870	0.491	0.211	0.809	0.667	0.616	0.036	0.781	0.626	0.528	0.044

A: sign accuracy; r: correlation; g: nonlinear gap; D: logit KL. Sign recovery is not independent computation.

Table 22: Depth profiling: all conditions, paired by code type.

α	K	Break E	Fraction E	Break R	Fraction R
0.050	2	3	0.045	3	0.045
0.050	4	3	0.045	3	0.045
0.050	8	3	0.045	3	0.045
0.050	16	3	0.045	3	0.045
0.150	2	3	0.045	3	0.045
0.150	4	3	0.045	3	0.045
0.150	8	3	0.045	3	0.045
0.150	16	3	0.045	3	0.045

E/R: extracted/random. Very large errors around injection require an indexing/numerical audit; these break statistics are not validated usable depths.

References

Hector Borobia, Elies Seguí-Mas, and Guillermina Tormo-Carbó. Near-floor geometry is generic: Leverage dispersion in trained overcomplete codes. arXiv:2605.01192v2, 2026. URL https://arxiv.org/abs/2605.01192v2.

Mouxiang Chen, Binyuan Hui, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Jianling Sun, Junyang Lin, and Zhongxin Liu. Parallel scaling law for language models. arXiv:2505.10475, 2025. URL https://arxiv.org/abs/2505.10475.

Minsoo Cheong, Donghyun Son, Woosang Lim, and Sungjoo Yoo. EntropyCache: Decoded token entropy guided KV caching for diffusion language models. arXiv:2603.18489v3, 2026. URL https://arxiv.org/abs/2603.18489v3.

Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. Transformer Circuits, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. arXiv:2312.00752, 2023. URL https://arxiv.org/abs/2312.00752.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning*, 2023. URL https://arxiv.org/abs/2301.10226.

Nicolas Menet, Michael Hersche, Geethan Karunaratne, Luca Benini, Abu Sebastian, and Abbas Rahimi. MIMONets: Multiple-input-multiple-output neural networks exploiting computation in superposition. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2312.02829.

Jose Joaquín Vargas Altalaguerri. Paper A: Activation-side session tagging for LLM audit logs. Companion manuscript, same repository; no DOI assigned, 2026a. URL https://github.com/ttzrs/neural-cdma/tree/papers-v2/papers/forensic.

Jose Joaquín Vargas Altalaguerri. Forensic provenance for LLM deployments: Real-time activation watermarking for legal non-repudiation. Working paper v1, Zenodo, 2026b. URL https://doi.org/10.5281/zenodo.20570543.

Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Huaijie Wang, Lingxiao Ma, Fan Yang, Ruiping Wang, Yi Wu, and Furu Wei. BitNet: Scaling 1-bit transformers for large language models. arXiv:2310.11453, 2023. URL https://arxiv.org/abs/2310.11453.

Zheyang Xiong, Ziyang Cai, John Cooper, Albert Ge, Vasilis Papageorgiou, Zack Sifakis, Angeliki Giannou, Ziqian Lin, Liu Yang, Saurabh Agarwal, Grigorios G. Chrysos, Samet Oymak, Kangwook Lee, and Dimitris Papailiopoulos. Everything everywhere all at once: LLMs can in-context learn multiple tasks in superposition. arXiv:2410.05603, 2024. URL https://arxiv.org/abs/2410.05603.